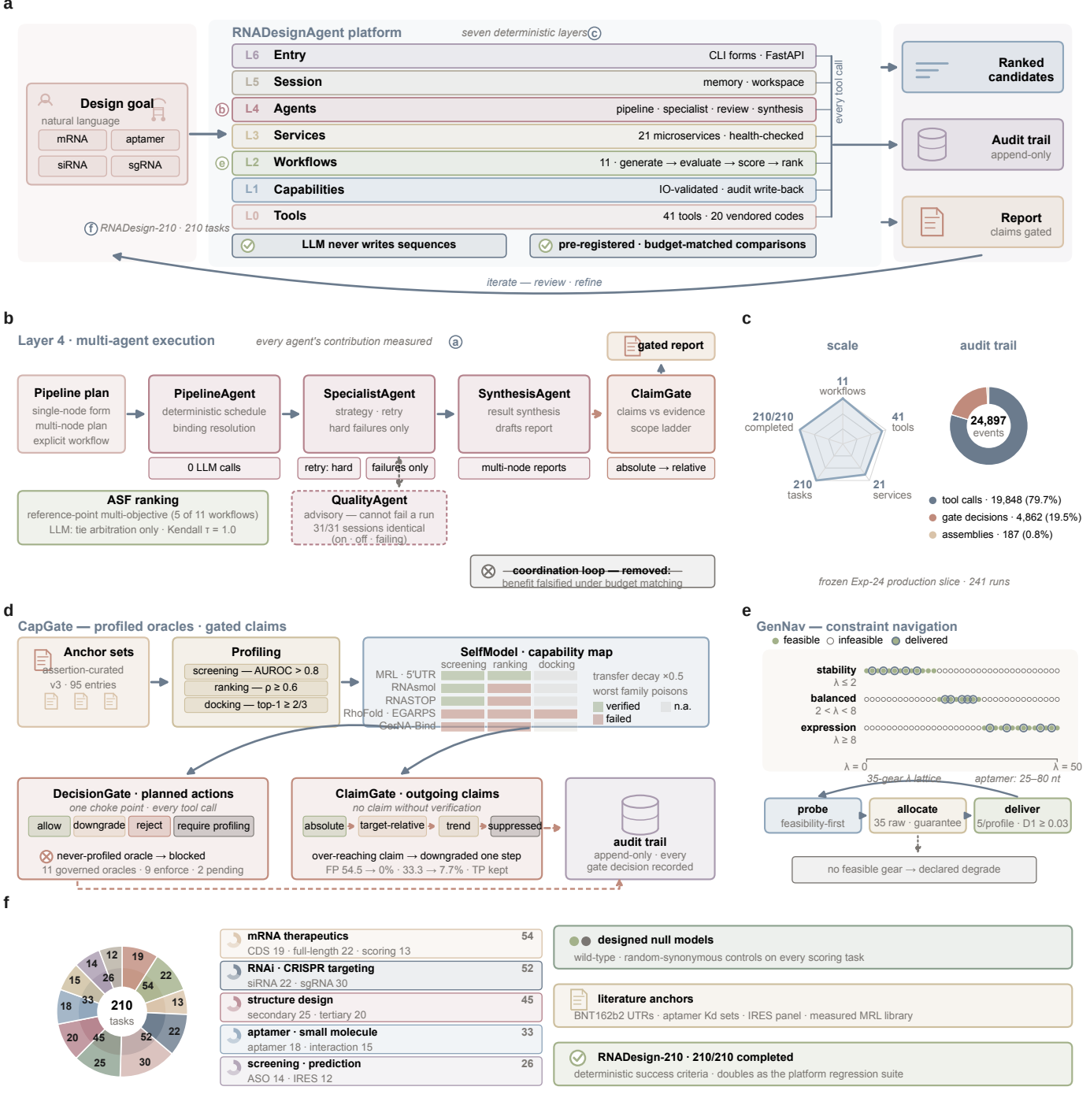

a
RNADesignAgent platform
seven deterministic layers
Design goal
natural language
mRNA
aptamer
siRNA
sgRNA
L6 Entry
CLI forms · FastAPI
L5 Session
memory · workspace
L4 Agents
pipeline · specialist · review · synthesis
L3 Services
21 microservices · health-checked
L2 Workflows
11 · generate → evaluate → score → rank
L1 Capabilities
IO-validated · audit write-back
L0 Tools
41 tools · 20 vendored codes
every tool call
RNADesign-210 · 210 tasks
LLM never writes sequences
pre-registered · budget-matched comparisons
Ranked candidates
Audit trail
append-only
Report
claims gated
iterate — review · refine
b
Layer 4 · multi-agent execution
every agent's contribution measured
Pipeline plan
single-node form
multi-node plan
explicit workflow
PipelineAgent
deterministic schedule
binding resolution
0 LLM calls
SpecialistAgent
strategy · retry
hard failures only
retry: hard
failures only
SynthesisAgent
result synthesis
drafts report
multi-node reports
ClaimGate
claims vs evidence
scope ladder
absolute → relative
gated report
ASF ranking
reference-point multi-objective (5 of 11 workflows)
LLM: tie arbitration only · Kendall τ = 1.0
QualityAgent
advisory — cannot fail a run
31/31 sessions identical
(on · off · failing)
coordination loop — removed:
benefit falsified under budget matching
c
scale
audit trail
11
workflows
41
tools
21
services
210
tasks
210/210
completed
24,897
events
tool calls · 19,848 (79.7%)
gate decisions · 4,862 (19.5%)
assemblies · 187 (0.8%)
frozen Exp-24 production slice · 241 runs
d
CapGate — profiled oracles · gated claims
Anchor sets
assertion-curated
v3 · 95 entries
Profiling
screening — AUROC > 0.8
ranking — ρ ≥ 0.6
docking — top-1 ≥ 2/3
SelfModel · capability map
screening
ranking
docking
MRL · 5′UTR
RNAsmol
RNASTOP
RhoFold · EGARPS
GenRNA-Bind
transfer decay ×0.5
worst family poisons
verified
failed
n.a.
DecisionGate · planned actions
one choke point · every tool call
allow
downgrade
reject
require profiling
never-profiled oracle → blocked
11 governed oracles · 9 enforce · 2 pending
ClaimGate · outgoing claims
no claim without verification
absolute
target-relative
trend
suppressed
over-reaching claim → downgraded one step
FP 54.5 → 0% · 33.3 → 7.7% · TP kept
audit trail
append-only · every
gate decision recorded
e
GenNav — constraint navigation
feasible
infeasible
delivered
stability
λ ≤ 2
balanced
2 < λ < 8
expression
λ ≥ 8
λ = 0
λ = 50
35-gear λ lattice
aptamer: 25–80 nt
probe
feasibility-first
allocate
35 raw · guarantee
deliver
5/profile · D1 ≥ 0.03
no feasible gear → declared degrade
f
210
tasks
mRNA therapeutics
CDS 19 · full-length 22 · scoring 13
54
RNAi · CRISPR targeting
siRNA 22 · sgRNA 30
52
structure design
secondary 25 · tertiary 20
45
aptamer · small molecule
aptamer 18 · interaction 15
33
screening · prediction
ASO 14 · IRES 12
26
designed null models
wild-type · random-synonymous controls on every scoring task
literature anchors
BNT162b2 UTRs · aptamer Kd sets · IRES panel · measured MRL library
RNADesign-210 · 210/210 completed
deterministic success criteria · doubles as the platform regression suite